Macroeconomics
ECON 101

Aggregate Expenditure and Output in the Short Run

(Chapter 8)

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Chapter Outline

8.1 The Aggregate Expenditure Model

8.2 Determining the Level of Aggregate Expenditure in the Economy

8.3 Graphing Macroeconomic Equilibrium

8.4 The Multiplier Effect

8.5 The Aggregate Demand Curve

Appendix The Algebra of Macroeconomic Equilibrium

The Aggregate Expenditure Model

8.1 Learning Objective

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

In this chapter, we will build up a simple mathematical model of the economy known as the *aggregate expenditure model*.

Aggregate expenditure model: A macroeconomic model that focuses on the short-run relationship between total spending and real GDP, assuming that the price level is constant.

Aggregate expenditure (AE): Total spending in the economy: the sum of consumption, planned investment, government purchases, and net exports.

This model will focus on short-run determination of total output in an economy

Four Components of Aggregate Expenditure

The four components in our model will be the same four that we introduced in a previous chapter as the components of GDP:

- ***Consumption (C)*:** Spending by households on goods and services.
- ***Planned investment (I)*:** Planned spending by firms on capital goods, and by households on new homes.
- ***Government purchases (G)*:** Spending on all levels of government on goods and services.
- ***Net exports (NX)*:** The value of exports minus the value of imports.

Aggregate expenditure is the sum of these components:

$$AE = C + I + G + NX$$

The Difference between Planned Investment and Actual Investment

Our aggregate expenditure model uses **planned investment,** rather than **actual investment;** in this way, the definition of aggregate expenditures is slightly different from GDP.

- The difference is that planned investment spending does not include the build-up of **inventories:** goods that have been produced but not yet sold:

Planned investment = Actual investment – unplanned change in inventories

Although Statistics Canada measures *actual investment*, we will assume that their measurement is close enough to *planned investment* to use in our estimates of aggregate expenditures.

Macroeconomic Equilibrium

Equilibrium in the economy occurs when spending on output is equal to the value of output produced; that is:

Planned aggregate expenditure = GDP

We know from previous chapters that GDP is generally changing from year to year; for simplicity, we will assume in this chapter that the economy is not growing.

Table 8.1 The Relationship Between Planned Aggregate Expenditure and GDP

If ...	then ...	and ...
planned aggregate expenditure equals GDP,	inventories don't change	the economy is in macroeconomic equilibrium.
planned aggregate expenditure is less than GDP,	inventories rise	GDP and employment decrease.
planned aggregate expenditure is greater than GDP,	inventories fall	GDP and employment increase.

- If planned expenditure is greater than GDP, firms have sold more than they expected to sell, inventories will decline and as a result of firms' reactions to the drop in inventories, GDP and total employment will increase.
- Similarly, when planned expenditure is less than GDP, inventories will increase, and GDP and total employment will decrease.

Table 8.2 Components of Aggregate Expenditure, 2021

Expenditure Category	Real Expenditure (millions of 2012 dollars)
Consumption	1 227 356
Planned Investment	378 117
Government Spending	530 077
Net Exports	–24 353

SOURCE: Statistics Canada, "Gross Domestic Product, Expenditure-Based, Canada, Quarterly (×1 000 000)," Table 36-10-0104-01.

The table below shows the values of the components of expenditure in 2021, with prices in 2012 dollars.

Clearly consumption is the largest portion, with government spending next followed by investment.

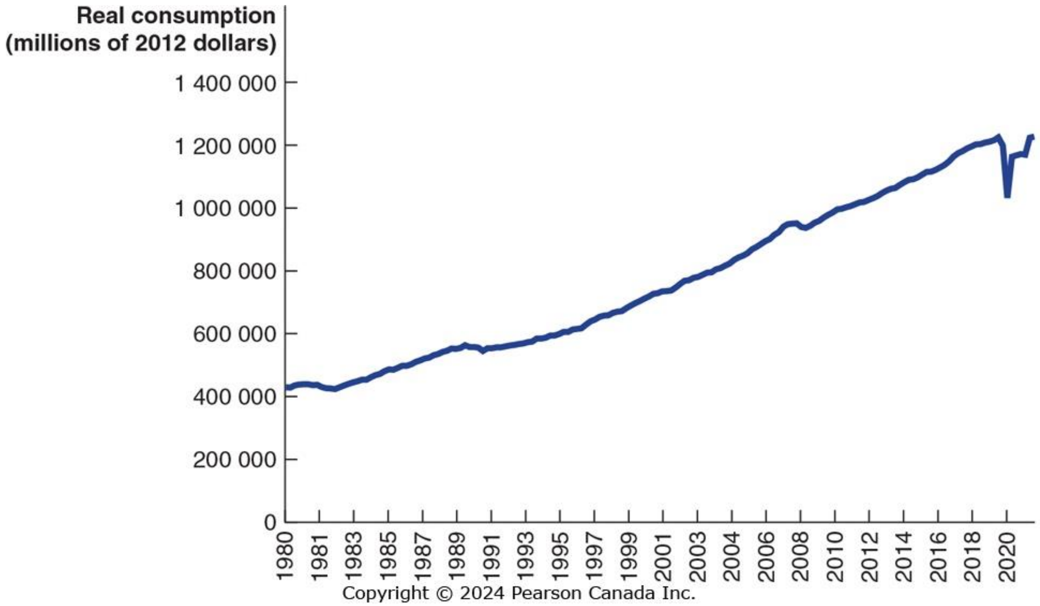
Net exports were negative in 2021: the value of Canadian imports was greater than the value of Canadian exports.

Figure 8.1 Real Consumption

Consumption tends to follow a relatively smooth, upward trend; its growth declines during periods of recession.

What affects the level of consumption?

1. **Current disposable income**
2. **Household wealth**
3. **Expected future income**
4. **The price level**
5. **The interest rate**



SOURCE: Statistics Canada, "Gross Domestic Product, Expenditure-Based, Canada, Quarterly (×1000 000)," Table 36-10-0104-01.

Determinants of Consumption (1 of 2)

Current disposable income

Consumer expenditure is largely determined by how much money consumers receive in a given year. We measure this by personal income, minus personal income taxes, plus government *transfer payments* such as Social Security.

Income in Canada expands in most years and so does consumption.

Household wealth

A household's *wealth* can be thought of as its *assets* (like homes, stocks and bonds, and bank accounts) minus its *liabilities* (mortgages, student loans, etc.).

Households with greater wealth will spend more on consumption, even with similar incomes. A recent estimate indicates that for a permanent $1 increase in household wealth, consumption spending increases by 4 to 5 cents per year.

Determinants of Consumption (2 of 2)

Expected future income

Most people prefer to keep their consumption fairly stable from year to year, a process known as *consumption-smoothing*.

Example: *Salespeople working on commission might have high incomes in some years, and low incomes in others. In order to predict their consumption, we would need to know what they believed their income would be in the future.*

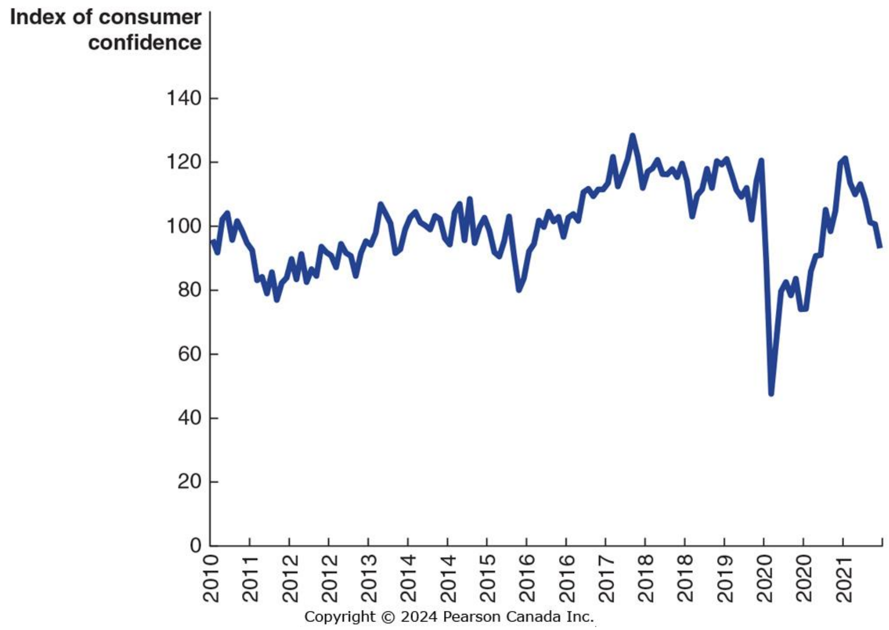
The price level

As prices rise, household wealth falls. If you have $100,000 in the bank, that will buy fewer products at higher prices. Consequently, higher prices result in lower consumption spending.

The interest rate

Higher real interest rates encourage saving rather than spending; so they result in lower spending, especially on *durable goods*.

Apply the Concept: Do Changes in Consumer Confidence Affect Consumption Spending

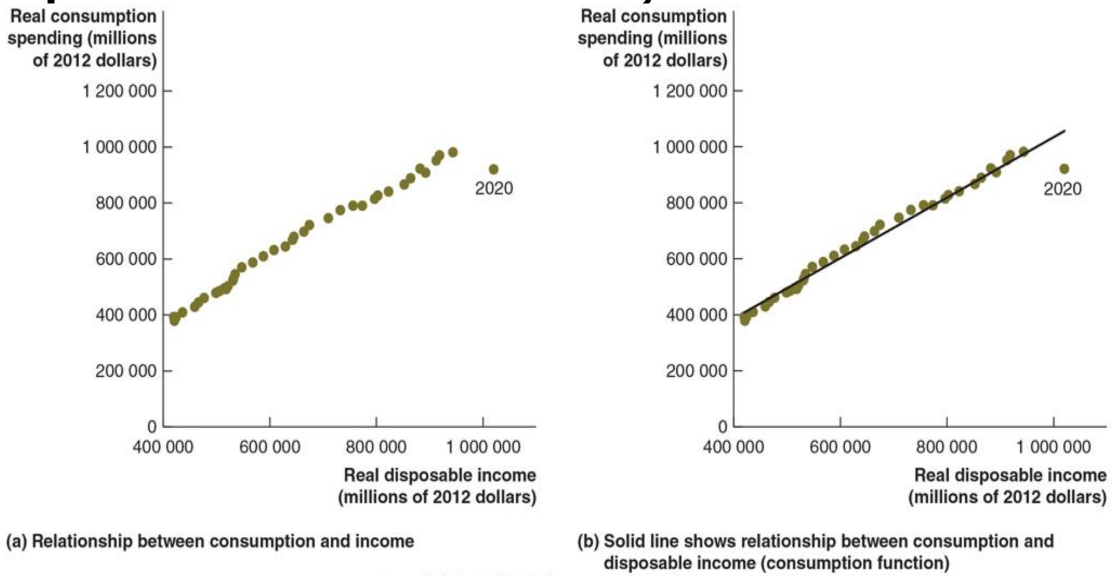


SOURCE: "Do Changes in Consumer Confidence Affect Consumption Spending?" The Conference Board of Canada, http://www.conferenceboard.ca/topics/economics/Consumer_confidence.aspx. Reprinted with permission.

Consumption spending depends on how people feel about the future. If people are confident, they spend more. If they're uncertain, they spend less. This helps predict consumption.

The Conference Board of Canada has been measuring ***Consumer Confidence Index*** monthly since 1980 to capture consumers' expectations about the future.

Figure 8.2 The Relationship Between Consumption and Income, 1981–2020



SOURCES: Statistics Canada. Table 36-10-0224-01 Household sector, current accounts, provincial and territorial, annual, https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3610022401.

How strong is the relationship between income and consumption?

As the graphs demonstrate, the answer is “very strong”. A straight line (called the **consumption function**) describes this relationship very well, suggesting that households spend a consistent fraction of each extra dollar of real disposable income on consumption.

Marginal Propensity to Consume

The graphs showed that consumers seem to have a relatively constant **marginal propensity to consume:** the amount by which consumption spending changes when disposable income changes.

This marginal propensity to consume (MPC) is the slope of the *consumption function*, the relationship between consumption spending and disposable income.

We can therefore estimate the MPC by estimating the slope of the production function:

$$MPC = \frac{\text{Change in consumption}}{\text{Change in disposable income}} = \frac{\Delta C}{\Delta YD}$$

For 2016–2017, we find:

$$\frac{\Delta C}{\Delta YD} = \frac{\$39.7\,\text{billion}}{\$51.9\,\text{billion}} = 0.764$$

So if incomes rose \$10 000, we estimate consumption would rise by \$10 000 × 0.764 = \$7640.

Consumption and National Income

The distinction between national income and GDP is relatively minor; for this simple model, we will assume they are equal, and use the terms interchangeably.

Since:

$$\text{Disposable income} = \text{National income} - \text{Net taxes}$$

where "net taxes" are equal to taxes minus transfer payments, we can write:

$$\text{National income} = \text{GDP} = \text{Disposable income} + \text{Net taxes}$$

If we assume that net taxes do not change as national income changes, we have the result that any change in disposable income is the same as the change in national income.

We will use this in the graph on the next slide.

Figure 8.3 The Relationship Between Consumption and National Income

The table shows the relationship between consumption and national income for an imaginary economy, keeping net taxes constant.

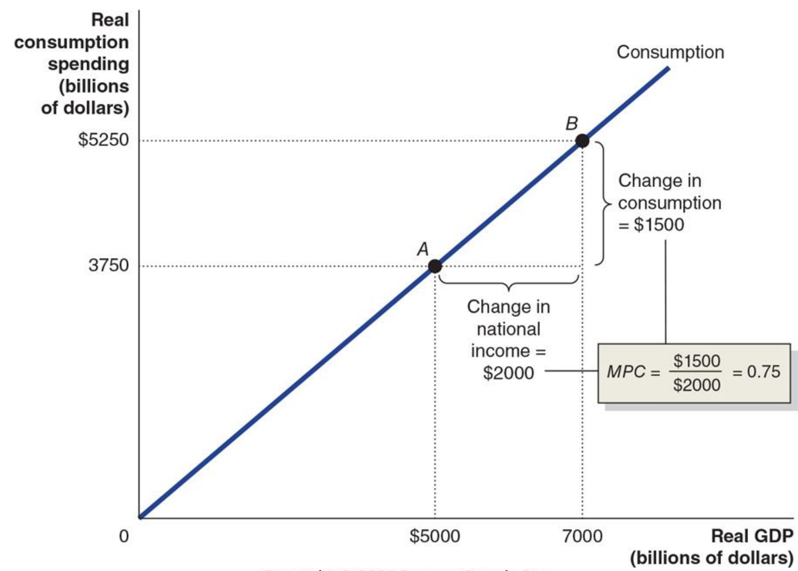
National Income or GDP (billions of dollars)	Net Taxes (billions of dollars)	Disposable Income (billions of dollars)	Consumption (billions of dollars)	Change in National Income (billions of dollars)	Change in Disposable Income (billions of dollars)
$ 1 000	$1 000	$ 0	$ 750	—	—
3 000	1 000	2 000	2 250	$2 000	$2 000
5 000	1 000	4 000	3 750	2 000	2 000
7 000	1 000	6 000	5 250	2 000	2 000
9 000	1 000	8 000	6 750	2 000	2 000
11 000	1 000	10 000	8 250	2 000	2 000
13 000	1 000	12 000	9 750	2 000	2 000

As national income rises by $2,000 billion…

… consumption rises by $1,500 billion.

So the marginal propensity to consume for this economy is:

$$MPC = \frac{\Delta C}{\Delta Y} = \frac{\$1{,}500 \text{ billion}}{\$2{,}000 \text{ billion}} = 0.75$$



Income, Consumption, and Saving

By definition, disposable income not spent is saved. Therefore we can write:

$$\text{National income} = \text{Consumption} + \text{Saving} + \text{Taxes}$$

$$Y = C + S + T$$

Any change in national income can be decomposed into changes in the items on the right hand side:

$$\Delta Y = \Delta C + \Delta S + \Delta T$$

We assume net taxes do not change, so $\Delta T = 0$; then:

$$\Delta Y = \Delta C + \Delta S$$

Now divide through by ΔY:

$$\frac{\Delta Y}{\Delta Y} = \frac{\Delta C}{\Delta Y} + \frac{\Delta S}{\Delta Y}$$

Marginal Propensity to Save

$$\frac{\Delta Y}{\Delta Y} = \frac{\Delta C}{\Delta Y} + \frac{\Delta S}{\Delta Y}$$

ΔS/ ΔY is the amount by which savings changes, when (disposable) income changes. This is known as the **marginal propensity to save**. We can rewrite the equation above as:

$$1 = MPC + MPS$$

That is, the marginal propensity to consume plus the marginal propensity to save must equal 1. This is because part of any increase in income is consumed, and the rest is saved.

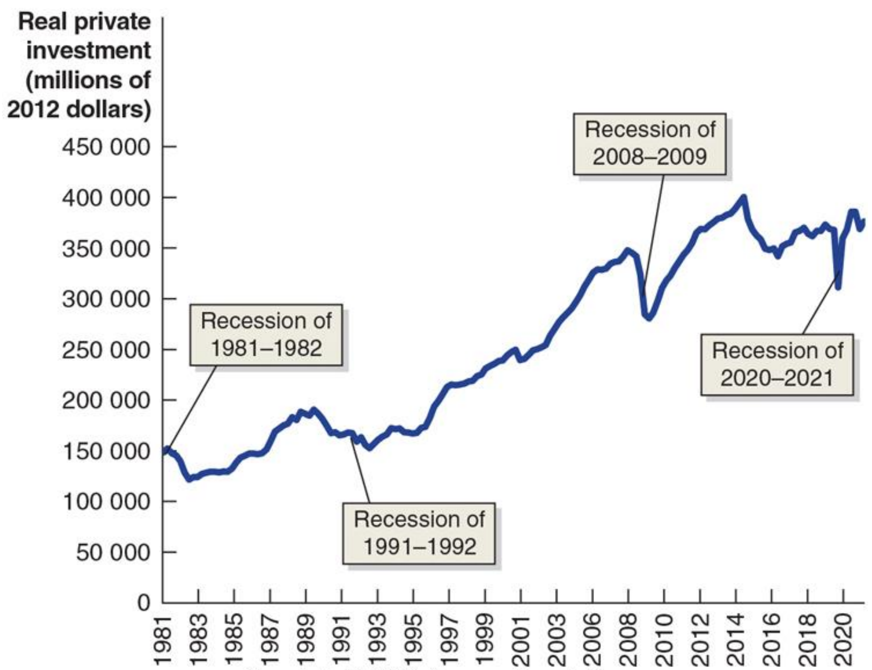
Figure 8.4 Real Investment

Investment has increased over time; but unlike consumption, it has not increased smoothly, and recessions decrease investment more.

What affects the level of investment?

1. **Expectations of future profit**
2. **Interest rate**
3. **Taxes**
4. **Cash flow**

We will examine how each of these affects the level of planned investment.



SOURCE: Statistics Canada, "Gross Domestic Product, Expenditure-Based, Canada, Quarterly (*1 000 000)," Table 36-10-0104-01.

Determinants of Planned Investment (1 of 2)

Expectations of future profits

Investment goods, such as factories, office buildings, machinery, and equipment, are long-lived. Firms build more of them when they are optimistic about future profitability. Recessions reduce confidence in future profitability, hence during recessions, firms reduce planned investment.

Purchases of new housing are included in planned investment. In recessions, households have reduced wealth, and less incentive to invest in new housing.

Interest rate

Since business investment is sometimes financed by borrowing, the real interest rate is an important consideration for investment.

A higher real interest rate results in less investment spending, and a lower real interest rate results in more investment spending.

Determinants of Planned Investment (2 of 2)

Taxes

Higher *corporate income taxes* on profits decrease the money available for reinvestment and decrease incentives to invest by diminishing the expected profitability of investment.

Similarly, *investment tax incentives* tend to increase investment.

Cash flow

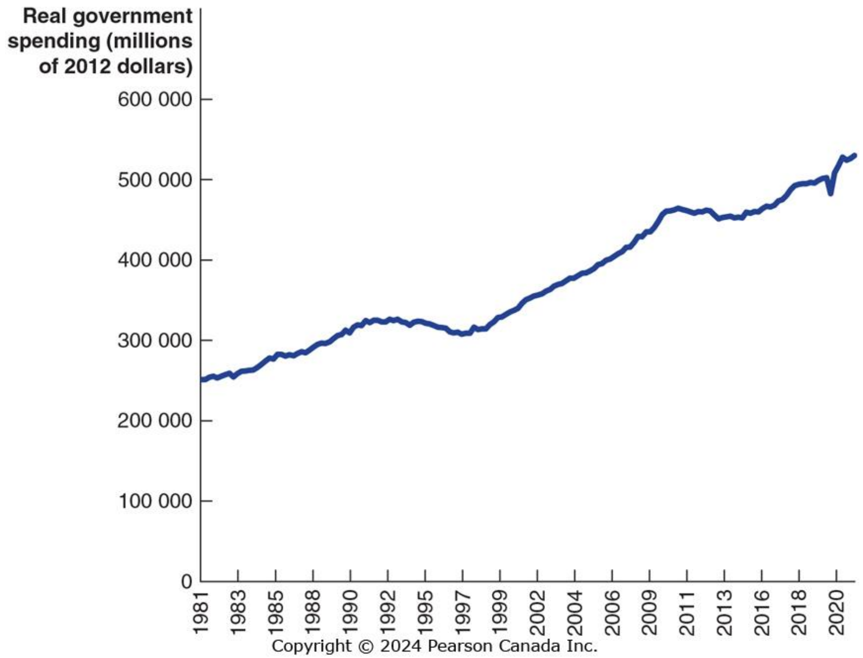
Firms often pay for investments out of their own **cash flow,** the difference between the cash revenues received by a firm and the cash spending by the firm.

The largest contributor to cash flow is profit. During recessions, profits fall for most firms, decreasing their ability to finance investment.

Figure 8.5 Real Government Purchases

Real government purchases include purchases at all levels of government: federal, provincial and local.

- This category does *not* include transfer payments; only purchases for which the government receives some good or service.



SOURCE: Statistics Canada, "Gross Domestic Product, Expenditure-Based, Canada, Quarterly, (×1000 000)," Table 36-10-0104-01.

Government purchases have generally, though not consistently, increased over time; exceptions include the early 1990s and during 2020 (COVID-19)

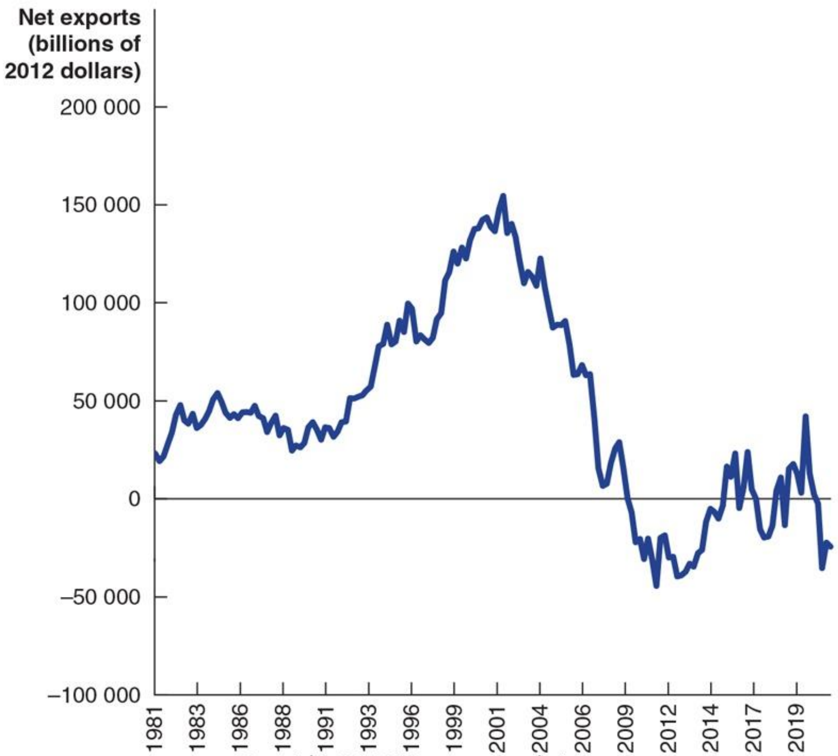
Figure 8.6 Real Net Exports

Net exports equals exports minus imports.

The value of net exports is affected by:

- *Price level in Canada vs. the price level in other countries*
- *Canadian growth rate vs. growth rate in other countries*
- *Canadian dollar exchange rate*

Net exports have been negative for over 10 years.



SOURCE: Statistics Canada, "Gross Domestic Product, Expenditure-Based, Canada, Quarterly, (×1000 000)," Table 36-10-0104-01.

Determinants of Net Exports

If…	Net Exports will…	…because…
…Canadian **price level** rises ***faster*** than foreign price levels…	decrease	Canadian goods become more expensive relative to foreign goods; so imports rise and exports fall.
…***slower***…	increase	The opposite is true.
…Canadian **GDP** grows ***faster*** than foreign GDP…	decrease	Canadian demand for imports rises faster than foreign demand for our exports.
…***slower***…	increase	The opposite is true.
…Canadian dollar ***rises*** in value relative to other currencies…	decrease	Imports are cheaper, and our exports are more expensive. So imports rise and exports fall.
…***falls***…	increase	The opposite is true.

Graphing Macroeconomic Equilibrium

8.3 Learning Objective

Use a 45°-line diagram to illustrate macroeconomic equilibrium.

Suppose in the whole economy, there is a single product: Coffee.

For the economy to be in equilibrium, the amount of coffee produced must equal the amount of coffee sold.

- Otherwise, inventories of Coffee rise or fall.

Figure 8.7 The 45°-Line Diagram

Any point on the 45° line could be an equilibrium—like points *A* or *B*.

At point *C*, the economy's inventories of coffee are being depleted, and production must rise.

At point *D*, inventories of coffee are growing, so production must fall.

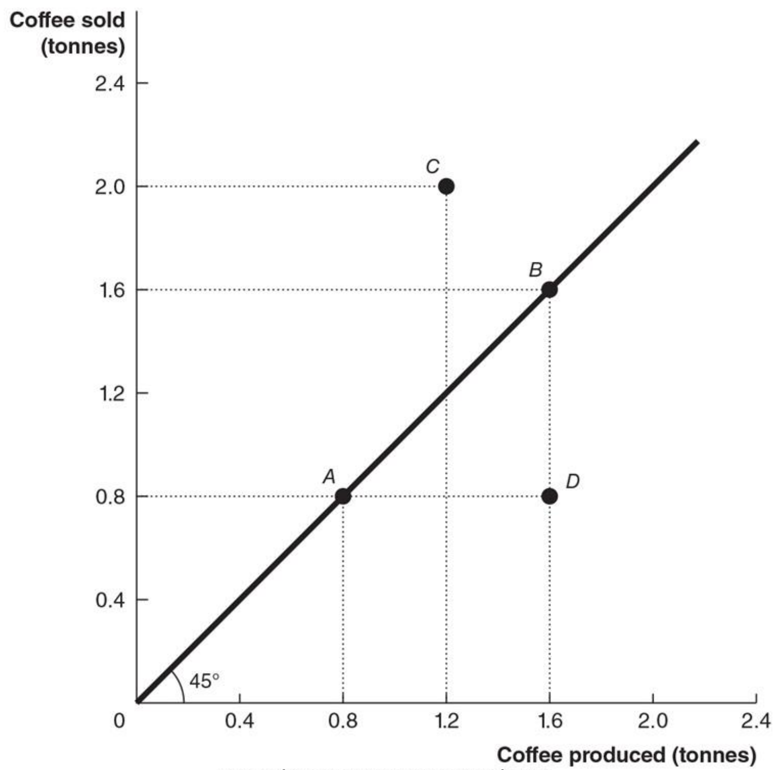
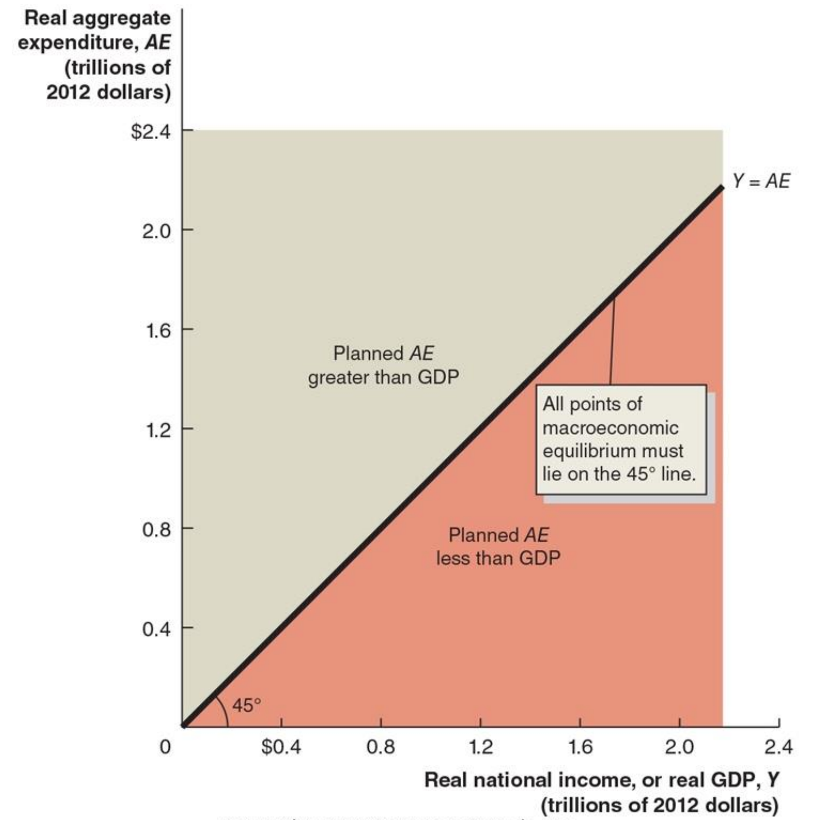


Figure 8.8 The Relationship Between Planned Aggregate Expenditure and GDP in a 45°-Line Diagram

We can apply this model to a real economy, with real national income (GDP) on the x-axis, and real aggregate expenditure on the y-axis.

This model is also known as the *Keynesian cross*, because it is based on the analysis of economist John Maynard Keynes.

Only points on the 45° line can be a macroeconomic equilibrium, with planned aggregate expenditure equal to GDP.



Determining the Macroeconomic Equilibrium

Any point on the 45° line *could* be an equilibrium; but how do we know which one *will* be the equilibrium in a given year?

- To determine this, recall that when they receive additional income, households consume some of it, and save some of it.
- The resulting *consumption function* tells us how much consumers will spend (real expenditure) when they have a particular income (real GDP).

This will determine Consumption (C) in the equation

$$Y = C + I + G + NX$$

Macroeconomic equilibrium simply means the left side (real GDP) must equal the right side (planned aggregate expenditure).

- The trick is to find the "right" level of C. For that, we use the 45° line diagram.

Figure 8.9 Macroeconomic Equilibrium on the 45°-Line Diagram

We start by building aggregate expenditure one element at a time in the diagram. We end up with AE (the pink line).

Macroeconomic Equilibrium is Income = Expenditure Y = C + I + G + NX.

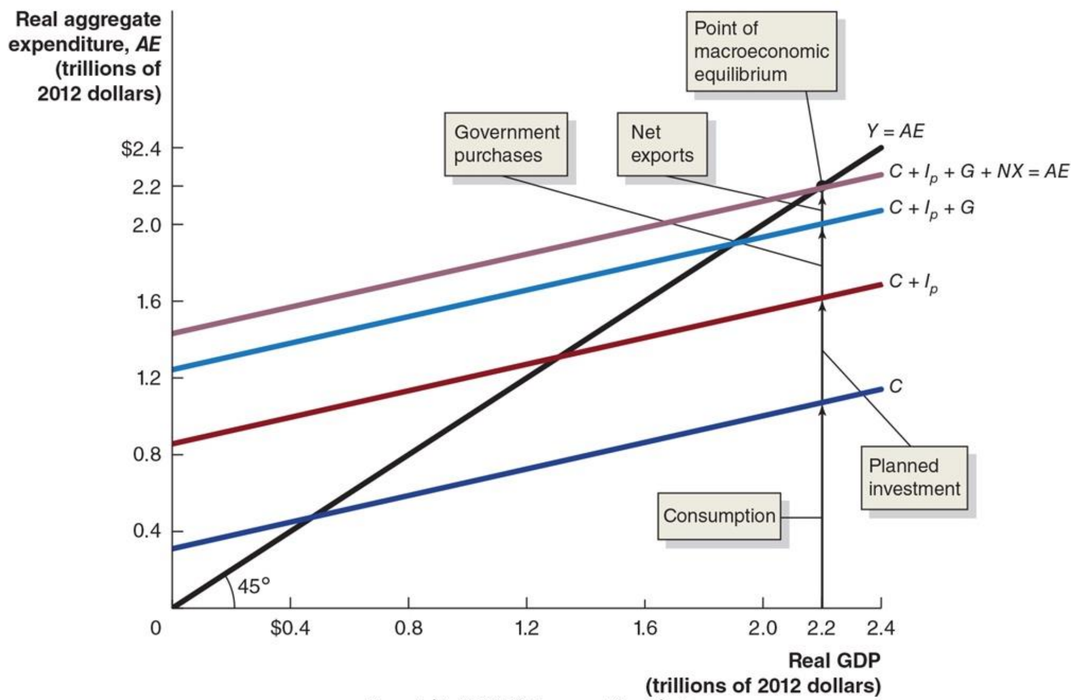


Figure 8.10 Macroeconomic Equilibrium

In this economy, macroeconomic equilibrium occurs at $2.2 trillion.

What if real GDP were lower, say $2.2 trillion?

- Aggregate expenditure would be higher than GDP, so inventories would fall.
- This would signal firms to increase production, increasing GDP.

The reverse would occur if real GDP were *above* $2.2 trillion.

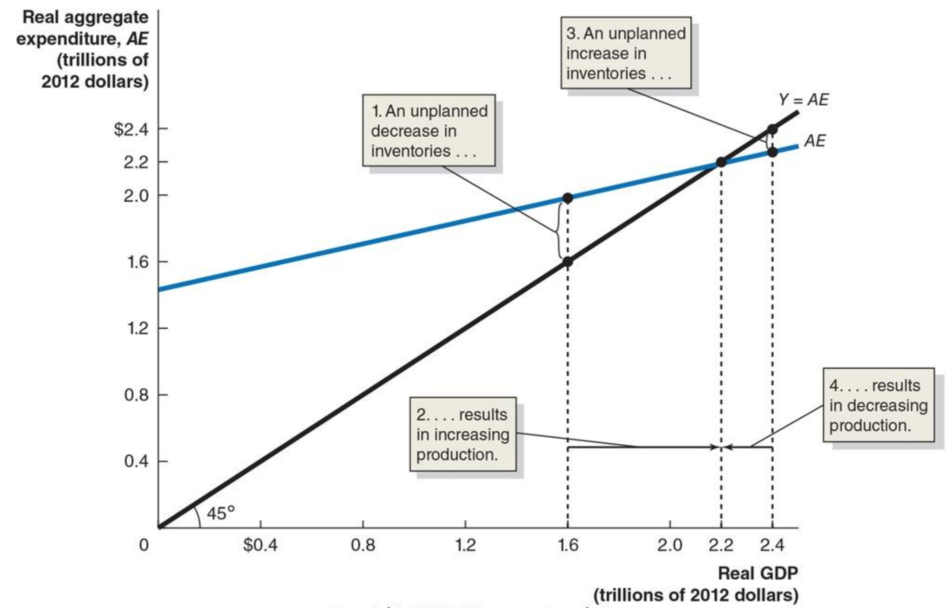


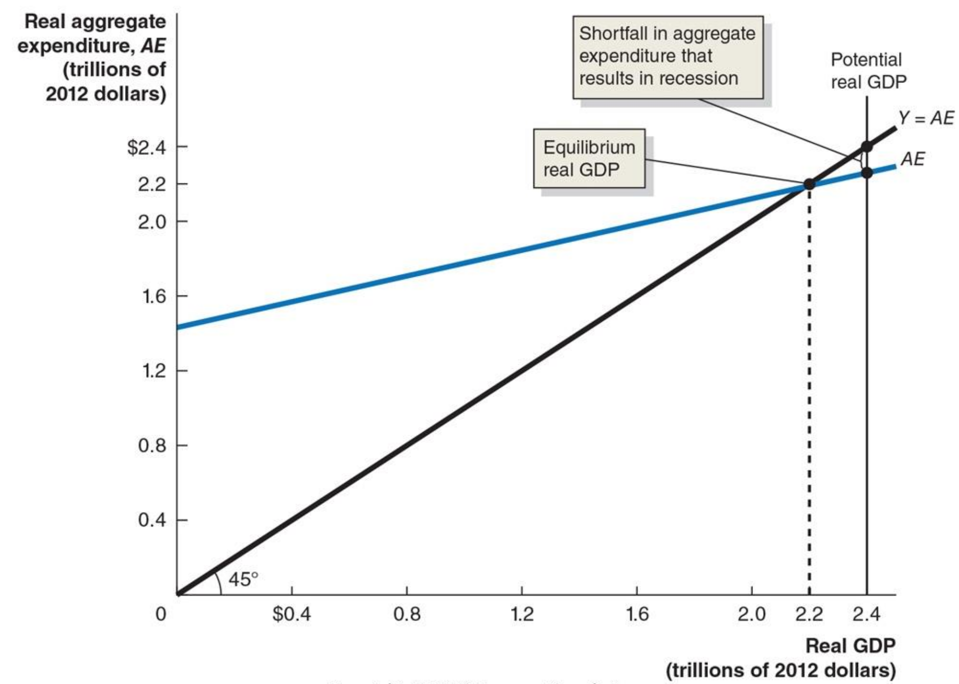
Figure 8.11 Showing a Recession on the 45°-Line Diagram

Macroeconomic equilibrium can occur anywhere on the 45° line. Ideally, we would like it to occur at the level of *potential GDP*.

If equilibrium occurs at this level, unemployment will be low—at the ***natural rate of unemployment,*** or the *full employment* level.

But for various reasons, this might not occur. For example, maybe firms are pessimistic and reduce investment spending.

- Then the equilibrium will occur *below* potential GDP—a recession.



The Important Role of Inventories

Inventories play a critical role in this model of the economy.

When planned aggregate expenditure is less than real GDP, firms will experience unplanned increases in inventories.

Then even if spending returns to normal levels, firms have excess inventories to sell before they can return to producing at normal levels.

Table 8.3 Macroeconomic Equilibrium

The table below shows several hypothetical combinations of real GDP and planned aggregate expenditure.

Real GDP (Y)	Consumption (C)	Planned Investment (Ip)	Government Purchases (G)	Net Exports (NX)	Planned Aggregate Expenditure (AE)	Unplanned Inventory Investment	Real GDP Will...
$800	$740	$125	$125	−$30	$960	−$160	increase
1200	1060	125	125	−30	1280	−80	increase
1600	1380	125	125	−30	1600	0	be in equilibrium
2000	1700	125	125	−30	1920	80	decrease
2400	2020	125	125	−30	2240	160	decrease

NOTE: The values are in billions of 2012 dollars.

Table 8.3 Macroeconomic Equilibrium

As real GDP changes, consumption changes but planned investment, government purchases, and net exports do not change with GDP.

Macroeconomic equilibrium can occur only at $1600 billion; otherwise, the unplanned change in inventories will cause firms to change production and real GDP will change.

The Multiplier Effect

8.4 Learning Objective

Describe the multiplier effect and use the multiplier formula to calculate changes in equilibrium GDP.

You may have noticed that a small change in planned aggregate expenditure causes a larger change in equilibrium real GDP.

In our model, planned investment, government purchases, and net exports are **autonomous expenditures:** expenditures that do not depend on the level of GDP.

- But, consumption has both an autonomous and **induced** effect.
- So, its level **does** depend on the level of GDP, and this produces the upward-sloping *AE* line.

Figure 8.12 The Multiplier Effect

An increase in an autonomous expenditure shifts the aggregate expenditure line upward.

When this happens, real GDP increases by *more* than the change in autonomous expenditures; this is the **multiplier effect.**

- The value of the increase in equilibrium real GDP divided by the increase in autonomous expenditures is the **multiplier.**

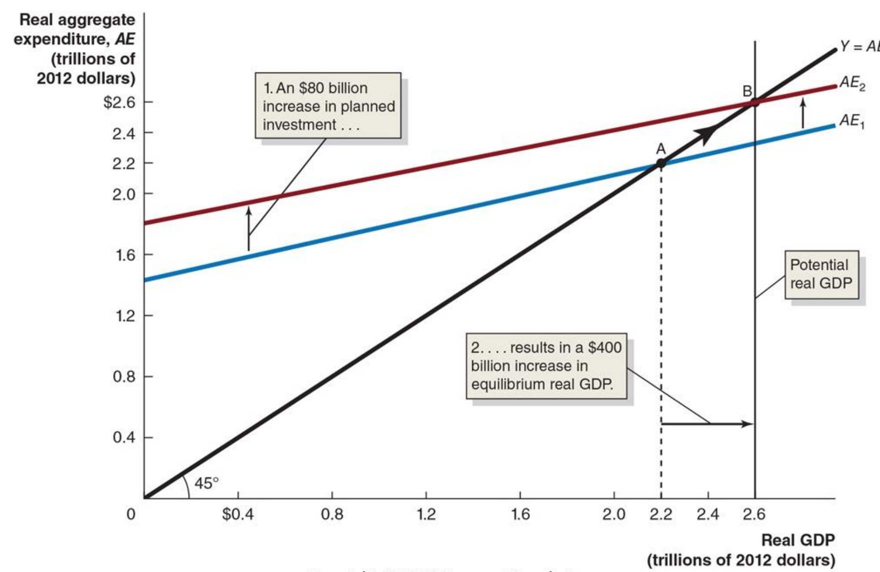


Table 8.4 The Multiplier Effect in Action

Initially, real GDP rises by the amount of the increase in autonomous expenditure.
This causes an increase in real GDP, which causes an increase in production, which causes an increase in real GDP…

	Additional Autonomous Expenditure	Additional Induced Expenditure (C)	Total Additional Expenditure = Total Additional GDP
Round 1	$80 billion	$0	$80.00 billion
Round 2	0	64.00 billion	144.00 billion
Round 3	0	51.20 billion	195.20 billion
Round 4	0	40.96 billion	236.16 billion
Round 5	0	32.77 billion	268.93 billion
.	.		.
.	.		.
.	.		.
Round 10	0	10.74 billion	357.05 billion
.	.		.
.	.		.
.	.		.
Round 20	.	1.15 billion	395.39 billion
.	.		.
Round n	0	0	400.00 billion

Eventual Effect of the Multiplier

We cannot say how long this adjustment to macroeconomic equilibrium will take—how many “rounds”, back and forth.

But we can calculate the value of the multiplier, as the eventual change in real GDP divided by the change in autonomous expenditures (planned investment, in this case):

$$\frac{\Delta Y}{\Delta I} = \frac{\text{Change in real GDP}}{\text{Change in investment spending}} = \frac{\$400 \text{ billion}}{\$80 \text{ billion}} = 5$$

With a multiplier of 5, each $1 increase in planned investment (or any other autonomous expenditure) eventually increases equilibrium real GDP by $5.

The Multiplier and the Marginal Propensity to Consume

How can we know the eventual value of the multiplier?

- In each "round", the additional income prompts households to consume some fraction (the *marginal propensity to consume*).

The total change in equilibrium real GDP equals:

The initial increase in planned investment spending	= *$80* billion
Plus the first induced increase in consumption	= $MPC \times \$80$ billion
Plus the second induced increase in consumption	= $MPC \times (MPC \times \$80$ billion)
	= $MPC^2 \times \$80$ billion
Plus the third induced increase in consumption	= $MPC \times (MPC^2 \times \$80$ billion)
	= $MPC^3 \times \$80$ billion
Plus the fourth induced increase in consumption	= $MPC \times (MPC^3 \times \$80$ billion)
	= $MPC^4 \times \$80$ billion
And so on ...	

A Formula for the Multiplier (1 of 2)

This becomes the infinite sum:

$$\text{Total change in GDP} = \$80\text{ billion} + MPC \times \$80\text{ billion} + MPC^2 \times \$80\text{ billion} + MPC^3 \times \$80\text{ billion} + MPC^4 \times \$80\text{ billion} + \ldots)$$

Which we can rewrite as:

$$\text{Total change in GDP} = \$80\text{ billion} \times (1 + MPC + MPC^2 + MPC^3 + MPC^4 + \ldots)$$

by factoring out the initial $80 billion increase in investment. Since MPC is less than 1, the expression in parentheses is:

$$\frac{1}{1 - MPC}$$

A Formula for the Multiplier (2 of 2)

In our case, MPC = 0.8; so the multiplier is 1/(1-0.8) = 5. A $80 billion increase in investment eventually results in a $400 billion increase in equilibrium real GDP.

The general formula for the multiplier is:

$$\text{Multiplier} = \frac{\text{Change in equilibrium real GDP}}{\text{Change in autonomous expenditure}} = \frac{1}{1 - MPC}$$

Summarizing the Multiplier Effect

1. The multiplier effect occurs both for an increase and a decrease in planned aggregate expenditure.
2. Because the multiplier is greater than 1, the economy is sensitive to changes in autonomous expenditure.
3. The larger the MPC, the larger the value of the multiplier.
4. Our model is somewhat simplified, omitting some real-world complications. For example, as real GDP changes, imports, inflation, interest rates, and income taxes will change.

The last point generally means that the value we estimate for the multiplier, from the MPC, is too high. In the next chapter, we will address some of these shortcomings.

The Paradox of Thrift

Recall the savings identity: *savings equals investment*.

- This implied that savings were the key to long-term growth.

But consider what happens in the short-term if people save more: consumption decreases, which causes aggregate expenditure to fall and… potentially pushes the economy into recession.

- John Maynard Keynes referred to this as the ***paradox of thrift*:** what appears to be favorable in the long-run may be counterproductive in the short-run.

Economists debate whether this paradox of thrift really exists; increasing savings decreases the real interest rate; the consequent increase in investment spending may offset the decrease in consumption spending.

- This is a real-world, data-driven debate, unable to be settled by our simple model.

The Aggregate Demand Curve

8.5 Learning Objective

Understand the relationship between the aggregate demand curve and aggregate expenditure.

The Price Level and Aggregate Expenditure

As demand for a product rises, two things are likely to occur: production increases, and so does the product's price.

- Our model has focused just on the real GDP, but what about changes in the price?

In a larger economy, we also expect that an increase in demand for everything (aggregate expenditure) would increase the price of everything (*price level*).

Will this price level change have a feedback-effect on aggregate expenditure?

- We generally expect that it will: increases in the price level will cause aggregate expenditure to fall, and decreases in the price level will cause aggregate expenditures to rise.

How does the Price Level Affect Aggregate Expenditure?

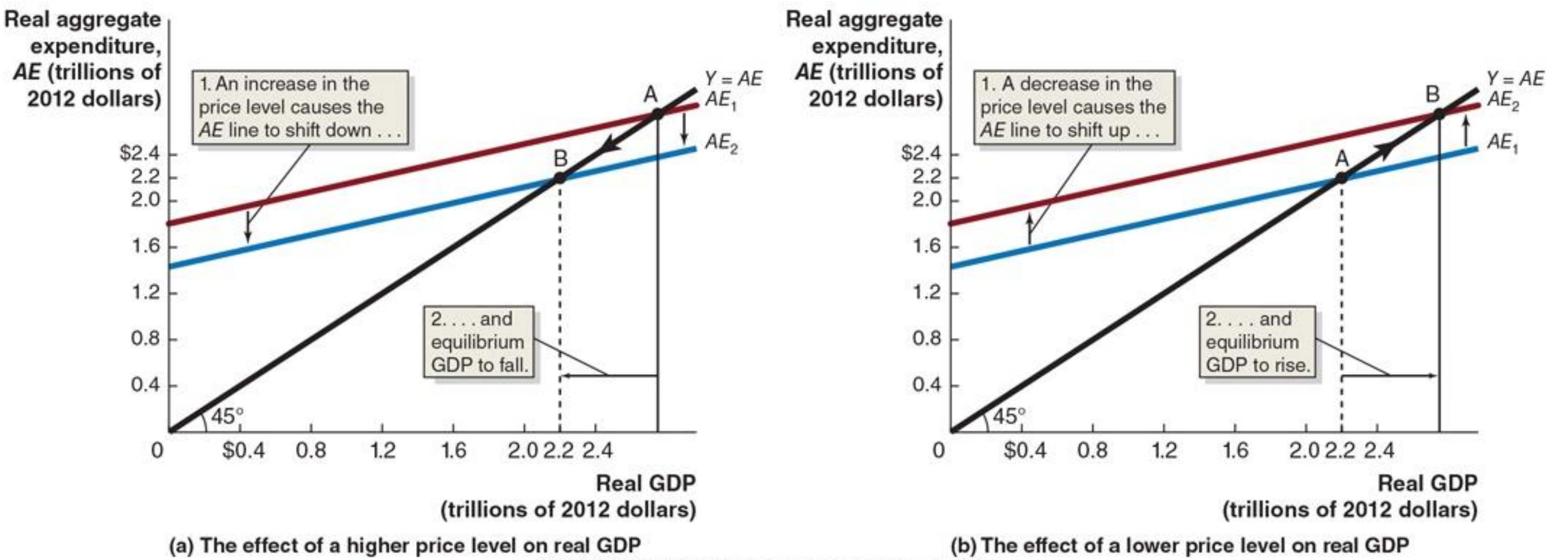
The price level affects aggregate expenditure in three ways:

1. Rising price levels decrease the real value of household wealth, causing consumption to fall.
2. If the price level in the Canada rises faster than in other countries, Canadian exports fall and imports rise, causing net exports to fall.
3. When prices rise, firms and households *need more money* to finance buying and selling. If the supply of money doesn’t change, the interest rate must rise; this will cause investment spending to fall.

A falling price level will have the opposite effect.

Each effect works in the same direction; so rising price levels decrease aggregate expenditure, while falling price levels increase aggregate expenditure.

Figure 8.13 The Effect of a Change in the Price Level on Real GDP



The diagrams show the effects described on the previous slide:

Panel (a): Increases in the price level cause *AE* and real GDP to fall.

Panel (b): Decreases in the price level cause *AE* and real GDP to rise.

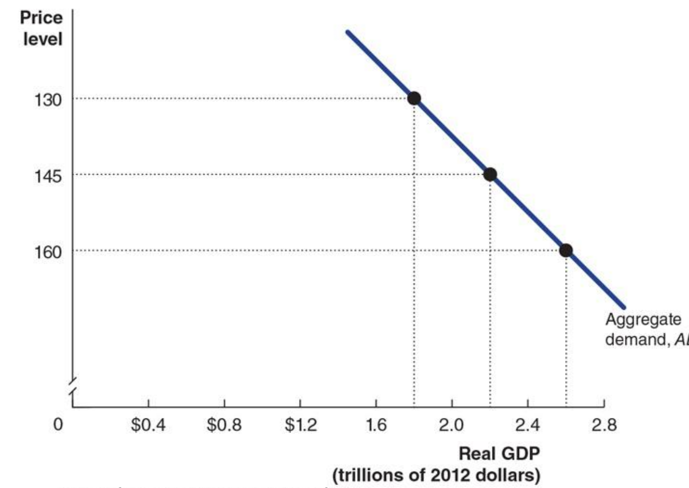
Figure 8.14 The Aggregate Demand Curve

Consequently, there is an inverse relationship between the price level and real GDP.

This relationship is known as the aggregate demand curve.

Aggregate demand (AD) curve: A curve that shows the relationship between the price level and the level of planned aggregate expenditure in the economy, holding constant all other factors that affect aggregate expenditure.

Price level	Equilibrium real GDP
130	$2.6 trillion
145	2.2 trillion
160	1.8 trillion



Common Misconceptions to Avoid

- In the 45°- line diagram, it becomes very important to distinguish income (real GDP) from expenditure (aggregate expenditure); although these are equal in macroeconomic equilibrium, they are not conceptually identical.
- Similarly, consumption spending is only a part (albeit the largest part) of aggregate expenditure.

Appendix The Algebra of Macroeconomic Equilibrium

Learning Objective

Apply the algebra of macroeconomic equilibrium.

Why Build a Numerical Model?

Graphical analysis of macroeconomic equilibrium can tell us the *qualitative changes* that take place.

- But an equation-based model can allow us to make *quantitative* or *numerical estimates* of what will occur.

Economists in universities, firms, and the government rely on *econometric models* in which they statistically estimate the relationships between economic variables.

Aggregate Expenditure Equations

Based on the example in the text, we can generate the following equations (changing the MPC so as to generate different results):

1. $C = 100 + 0.8Y$	Consumption function
2. $I_P = 125$	Planned investment function
3. $G = 125$	Government spending function
4. $NX = -30$	Net export function
5. $Y = C + I + G + NX$	Equilibrium condition

In using the model, researchers would estimate the *parameters* of the model—like the MPC or the values of the autonomous expenditure components like planned investment—using statistical methods and years of observations of data.

Solving the Model

The first four equations can be used to form the aggregate expenditure function—the right hand side of the fifth equation.

The fifth equation is the essential “equilibrium condition”, representing the effect of the 45°-line.

Substituting the first four equations into the fifth gives:

$$Y = 100 + 0.8Y + 125 + 125 - 30$$

Subtracting $0.65Y$ from both sides gives:

$$Y - 0.8Y = 100 + 125 + 125 - 30$$

Which simplifies to:

$$0.2Y = 320$$

$$Y = \frac{320}{0.2} = 1600$$

General Aggregate Expenditure Equations

More generally, we could allow the parameters of the model to be represented by letters.

1. $C = \overline{C} + MPC(Y)$	Consumption function
2. $IP = \bar{I}_p$	Planned investment function
3. $G = \overline{G}$	Government spending function
4. $NX = \overline{NX}$	Net export function
5. $Y = C + IP + G + NX$	Equilibrium condition

The letters with bars over them are parameters—fixed (autonomous) values. For example, $\overline{C}$ was 100 in our example.

Solving the General Aggregate Expenditure Equations

Solving now for equilibrium, we get

$$Y = \overline{C} + MPC(Y) + \overline{I} + \overline{G} + \overline{NX}$$

$$Y - MPC(Y) = \overline{C} + \overline{I} + \overline{G} + \overline{NX}$$

$$Y(1 - MPC) = \overline{C} + \overline{I} + \overline{G} + \overline{NX}$$

$$Y = \frac{\overline{C} + \overline{I} + \overline{G} + \overline{NX}}{1 - MPC} = (\overline{C} + \overline{I} + \overline{G} + \overline{NX}) \times \frac{1}{1 - MPC}$$

The last equation makes clear that:

Equilibrium GDP = Autonomous expenditure × Multiplier